NATIONAL INSTITUTE OF TECHNOLOGY, TRICHY

National Cyber Incident Analytics Platform (NCRI)

for Trend Analysis and Risk Assessment

MCA PROJECT REVIEW II

PRESENTED BY

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The Problem and the Platform's Purpose

Unifying fragmented cyber incident data for consistent, actionable insight.

Cross-Dimensional Comparison

Compare incidents across time, geography, sectors, and attack types.

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Support analysts, researchers, and policy-oriented decision makers.

Delivers context-rich situational intelligence tailored for diverse audiences—from tactical security analysts investigating active breaches to institutional leaders setting national cyber defense priorities.

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Provenanced & Repeatable Scoring

Provide traceable source provenance and repeatable risk scoring.

Ensures every recorded incident retains verifiable lineage back to authoritative reporting feeds. Calculates an explainable, deterministic National Cyber Risk Index (NCRI) rather than opaque estimations.

Data Sources and Collection Methods

Multiple trusted inputs. Unified intake. Consistent context.

HISTORICAL GLOBAL INCIDENTS

UMD CISSM

- **Feed: Historical global incident CSV export**

Extensive academic repository tracking cyber events globally across public, defense, and enterprise sectors.

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CISA KEV

- **Feed: Official JSON catalog; custom CSV/JSON imports**

Authoritative catalog of known exploited vulnerabilities linked conditionally when evidence confirms active weaponization.

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CENTRAL INTAKE

Unified Data Intake

Structured, standardized, and ready for analysis.

✓ **Universal Adapters**
Custom connector interfaces normalize diverse formats (REST, CSV, JSON).

✓ **Taxonomy Mapping**
Translates non-uniform sector, attack, and geography labels to ISO standards.

✓ **Provenance Tracking**
Every record maintains source links, ingestion batch IDs, and audit flags.

✓ **Ready for Analytics**
Cleaned datasets flow directly to persistence and index calculation engines.

What Data Is Captured?

Standardized schema capturing incident context, operational attributes, and linked CVE telemetry.

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CAPTURE SCOPE & PRINCIPLES

Incident Context

Incident date, victim organization, source provenance, and public reference URLs.

Taxonomic Attribution

Sector/subsector classifications, country/region ISO codes, attack type and subtype.

Impact & Actor Profile

Threat actor and motive when available; severity and operational impact where supported.

Strict Evidence-Based Linking

CVE metadata is stored separately and linked only when empirical evidence supports a direct relationship.

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CANONICAL SCHEMA FIELDS (10 DIMENSIONS)

Incident Date
ISO YYYY-MM-DD standardized date

Victim Organization
Cleaned entity / enterprise identifier

Source
Originating feed (CISSM, Ransomware.live, CERT-In)

Public Reference
Verifiable external citation URL

Sector / Subsector
Harmonized critical infrastructure taxonomy

Country / Region
ISO 3166-1 alpha-2 / alpha-3 standards

Attack Type / Subtype
Normalized attack classification

Threat Actor / Motive
Attribution & intent indicators (if available)

Severity / Impact
Normalized comparative impact metric

CVE Metadata
Linked known exploited vulnerability IDs

The Data Engineering Pipeline

Automated multi-stage pipeline ensuring schema harmonization, record uniqueness, and auditability.

STAGE 01

SOURCE ADAPTERS

Extract incident data from diverse sources

TEMPORAL NORMALIZATION

Date Standardization

Dates standardized to ISO YYYY-MM-DD across all incoming records. Resolves varied epoch formats, month-name text, and missing granular timestamps to ensure clean chronological aggregation.

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Map and standardize to common taxonomies

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SCHEMA HARMONIZATION

Taxonomy Alignment

Sectors, attack types, country names/codes, and severity fields normalized. Resolves alias mappings (e.g., 'Finance' vs. 'Banking') to a single unified reference model.

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SHA-256 RESOLUTION

Composite Fingerprint

SHA-256 composite fingerprint based on cleaned victim name, incident date, and attack type. Prevents identical breach events reported across multiple public feeds from inflating counts.

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INGESTION AUDIT

Process outcomes tracked for quality and transparency

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TRACEABLE PROVENANCE

Ingestion Audit Logging

Ingestion audit captures processed, added, skipped, updated, and failed records. Maintains an immutable log of pipeline execution runs for complete transparency and debugging.

System Architecture

Modular layered architecture connecting data ingestion, persistent storage, analytics modeling, and client presentation.

STAGE 01

DATA INGESTION & PIPELINE

Data Acquisition Layer

- Ransomware.live (REST API Feed)
- UMD CISSM Database (Global CSV Export)
- CISA KEV (Known Exploited JSON Catalog)
- CERT-In Disclosures (National Curated CSV)

▼ Automated Data Flow

Ingestion & Pipeline Layer

- BaseAdapter Interface (Universal Extractor)
- Normalization Engine (Taxonomy Harmonizer)
- Deduplication & SHA-256 Fingerprinting
- Ingestion Run Auditor (Batch Run Tracking)

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STAGE 02

STORAGE & RISK ENGINE

Persistence & Storage Layer

- PostgreSQL 15+ Relational Database Engine
- Tables: ingestion_runs | data_sources
- Tables: cyber_incidents | vulnerabilities
- Junctions: incident_vulnerabilities | risk_assessments

▼ Automated Data Flow

Analytics & Risk Engine Layer

- AnalyticsService (Core Engine Orchestrator)
- NCRI Engine (4-Weighted Composite Scoring)
- Predictive Analytics & Forecasting Models
- Sectoral & Temporal Aggregation Workers

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APPLICATION & PRESENTATION

REST API Application Layer

- FastAPI Asynchronous Web Framework
- Pydantic v2 (Strict Schema Validation)
- SQLAlchemy 2.0 ORM & Connection Pools
- Modular Endpoints (/incidents, /risk, /sources)

▼ Automated Data Flow

Presentation & Frontend Layer

- Next.js 14 App Router | React 18 Framework
- TypeScript | Tailwind CSS Design Tokens
- Recharts Dynamic Analytics & Trend Visuals
- Interactive Situational Dashboard (Port 3000)

How Data Becomes Insight

Standardized incident records flow through analytics to reveal patterns, context, and action.

DATA MATURATION STAGES

STAGE 1 - Standardized Incident Records

Cleaned raw records parsed across feeds with unified temporal, geographic, and attack-type attributes.

STAGE 2 - Aggregated & Normalized Data

Sector-level groupings, time-series binning, and severity normalizations structured for analytical modeling.

STAGE 3 - Analytics Engine & Models

Mathematical computation of weighted risk indices, rolling trendlines, and anomaly detection algorithms.

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KEV Catalog & Export
Search the CISA Known Exploited Vulnerabilities catalog and export tailored datasets for downstream research.

National Cyber Risk Index (NCRI)

An explainable 0–100 composite score compares observed risk across sectors using four weighted dimensions.

NCRI

0–100 Score

Sector-level comparative prioritization benchmark calculated deterministically.

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Log-scaled sector incident volume

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STANDARDIZED RISK TIERS (0–100 SCALE)

< 25
Low
Baseline risk level; routine defense.

25 – 49.9
Moderate
Standard monitoring; elevated frequency.

50 – 69.9
Elevated
Enhanced scrutiny; critical alert.

70 – 84.9
High
High operational disruption; active response.

85 – 100
Severe
Catastrophic impact or mass velocity.

Data Quality, Limitations, and Responsible Interpretation

Maintaining analytical integrity requires explicit recognition of data biases and boundary conditions.

Reporting Asymmetry & Incompleteness

Publicly reported incidents are incomplete and may be uneven across countries and sectors.

Mandatory breach reporting varies significantly across jurisdictions. Public disclosures capture only a fraction of overall activity, meaning volume metrics reflect reporting habits alongside true threat posture.

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Preserving Nulls vs. Data Fabrication

Financial loss may often be missing; unknown values should remain null rather than be fabricated.

Loss numbers are rarely disclosed in early reports. Imputing financial values or assuming averages creates deceptive precision; preserving missing fields guarantees analytical auditability.

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Comparative Index vs. Predictive Certainty

A high NCRI score is a comparative indicator under the defined methodology, not a probability of future attack.

The NCRI quantifies observed historical activity, baseline criticality, and exploitation traits for relative resource prioritization. It does not predict imminent breaches for individual entities.

Outputs and Next Steps

An auditable, extensible analytics workflow that enables investigation and comparison.

CURRENT DELIVERABLES

WHAT YOU GET

Interactive Analytics Dashboard

Full incident-level filtering by timeframe, geographic region, victim sector, and severity metrics.

National Risk Index (NCRI)

Automated 0–100 composite scoring engine for comparative sector risk evaluation and prioritization.

Source Registry & Ingestion Audit Logs

Complete operational tracking capturing processed, added, skipped, and updated records per ingestion run.

Exportable Standardized Datasets

Downloadable clean incident data in structured formats for academic research and secondary analysis.

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FUTURE ENHANCEMENTS

WHAT'S NEXT

Source Freshness Monitoring
Automated heartbeat alerts and health monitors detecting stale external feeds or unexpected API outages.

Validation of Incident-CVE Links
Advanced corroboration pipelines cross-referencing vendor advisories and CVSS metrics for high-confidence linkage.

Improved Coverage Assessment
Quantitative statistical modeling to evaluate under-reporting ratios across distinct geographic regions.

Formal Evaluation of Forecast Accuracy
Empirical backtesting and error-rate evaluation for predictive threat trajectory models.

THANK YOU